



11532 - It's the Little Things that Count: Unveiling Faint Merger Signatures and Tiny Companions around Massive Quiescent Galaxies at $z \sim 2$

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) SUSPENSE

ABSTRACT

The inside-out model of galaxy assembly, in which galaxies grow via many minor mergers after quenching, can explain many of their observed properties. However, we have lacked direct observational evidence of these mergers. Deep JWST imaging has finally revealed low-mass companion galaxies with average stellar masses 900x lower than their quiescent hosts. These tiny satellites may be responsible for decreasing rotation, strengthening colour and metal gradients, and tripling sizes of quiescent galaxies over the past 10 billion years. However, the initial JWST study has only shallow spectroscopy of a small fraction of the sample. To robustly test the inside-out model, we propose to obtain deep F150W2/F322W2 NIRCam imaging of 12 spectroscopically confirmed, massive ($M^* > 10^{11} M_{\text{sun}}$) quiescent galaxies at $z \sim 2$ from the public JWST-SUSPENSE survey. We will exploit the extra-wide filters to detect tiny companions and, for the first time at $z > 1$, extended tidal features signifying recent or ongoing merging. We will leverage a statistical method to measure average local overdensities of tiny companions around massive hosts. With this data and the exceptional SUSPENSE spectra we will (1) assess if all massive quiescent galaxies have a significant number of companions, validating cosmological simulation predictions, and (2) evaluate if $z \sim 2$ merger signatures (companions and tidal features) can truly explain the inside-out model. These deep NIRCam images will provide the first census of faint signatures of mergers around spectroscopically confirmed massive quiescent galaxies at $z \sim 2$, clarifying size, rotation, and stellar population evolution of quiescent galaxies over cosmic time.

OBSERVING DESCRIPTION

We propose to exploit the extra-wide F150W2/F322W2 filter configuration to obtain ultra-deep NIRCam imaging of 12 massive quiescent galaxies at $1.2 < z < 2.7$. These galaxies can be observed in just 1 NIRCam pointing.

To maximize the area with full depth, we will use the tight INTRAMODULEBOX dither pattern. Using the JWST ETC, with 10 groups per integration, 3 integrations per exposure, and 8 dithers, for an exposure time of 12798.23s, our estimated signal-to-noise limits are as follows:

- F150W2: S/N ~ 5 per pixel (for a tidal feature at $z \sim 2.7$ with a surface brightness of 27.8 magnitudes per square arcsecond) to S/N ~ 31 per pixel (for a tidal feature at $z \sim 1.6$ with a surface brightness of 25.8 magnitudes per square arcsecond)
- F322W2: S/N ~ 10 per pixel (for a tidal feature at $z \sim 1.3$ with a surface brightness of 27.9 magnitudes per square arcsecond) to S/N ~ 63 per pixel (for a tidal feature at $z \sim 1.6$ with a surface brightness of 25.8 magnitudes per square arcsecond).

Our field can be observed for a limited PA range. All 12 targets can be observed with limited V3PA ranges of 98.7-116.06 degrees and 284.96-299.4

JWST Proposal 11532 (Created: Tuesday, May 26, 2026, 8:00:40AM Eastern Standard Time) - Overview
degrees.

Proposal 11532 - Targets - It's the Little Things that Count: Unveiling Faint Merger Signatures and Tiny Companions around Massive ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SUSPENSE	RA: 10 02 1.0176 (150.5042400d) Dec: +02 27 52.81 (2.46467d) Equinox: J2000		
Comments: Category=Galaxy Description=[Compact galaxies, Dwarf galaxies, E+A galaxies, Giant elliptical galaxies] Extended=YES					

Proposal 11532 - Observation 1 - It's the Little Things that Count: Unveiling Faint Merger Signatures and Tiny Companions around Ma...

Observation	Proposal 11532, Observation 1										Tue May 26 13:00:40 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Observation 1) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(Visit 1:1) Warning (Form): Data Excess over lower threshold										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SUSPENSE	RA: 10 02 1.0176 (150.5042400d)								
			Dec: +02 27 52.81 (2.46467d)								
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Template	Module		Subarray				Target Placement				
	ALL		FULL				Module gap (large extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		8		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	SHALLOW4	10	3	24	8	12798.23		
Special Requirements	Aperture PA Range 108.01542306 to 115.98542306 Degrees (V3 108.09 to 116.06)										
	Aperture PA Range 293.00542306 to 299.32542306 Degrees (V3 293.08 to 299.4)										